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**An EEGNet-based approach to Motor Imagery
classification for Brain-Computer Interface
Systems**

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Abstract

Brain–Computer Interface (BCI) systems enable direct communication between the brain and external devices by translating neural signals into control commands. These systems can decode users’ imagined movements (motor imagery) and use them to operate assistive technologies and prosthetics. This research focuses on developing a compact, generalisable, and high-accuracy model for motor imagery classification.

A comprehensive signal pre-processing pipeline is designed, improving input signal quality and model performance. A convolutional neural network (inspired by EEGNet [1]) is proposed to classify processed signals into predicted motor imagery output. Targeted architectural enhancements improve network robustness and representational capacity. The model’s performance is evaluated using both same-subject and cross-subject evaluation methodologies, assessed on two benchmark datasets: BCI Competition IV Datasets 2a and 2b [2, 3].

The proposed model consistently outperforms baseline models on Dataset 2a, achieving averages of 65.70% same-subject accuracy, and 45.29% cross-subject accuracy. These results show how targeted and strategic enhancements to even established classifiers can deliver notable improvements in accuracy, contributing to the advancement of generalisable EEG classification methods.

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CHAPTER 1

Background

Brain-Computer Interface (BCI) systems enable direct communication between the brain and external devices through the translation of neural signals into motor commands [4]. The emergence of these devices has created transformative possibilities for assistive technologies and prosthetics, particularly for utilisation by amputees and the physically disabled [5]. Electroencephalography (EEG), a non-invasive method for recording neuronal impulse activity, has been harnessed to capture motor imagery (MI) data used to train and operate such systems [6, 7, 8, 1]. This technology is of great potential impact, as it promises to offer individuals with severe motor impairments a means of regaining autonomy and interacting with their environment, even without the capacity for physical movement.

1.1 Brain-Computer Interfaces

1.1.1 Introduction to BCIs

Brain-Computer Interfaces (BCIs) facilitate communication by taking inputs in the form of brain-originated electrical impulses and passing machine-readable commands as output. In doing so, these devices enable communication without reliance on typical pathways of peripheral nerves and muscles [9]. These are powerful technologies with significant potential across a wide range of applications, offering a number of solutions across medical domains [4]. In particular, these systems have immense capacity for application in prosthesis and assistance of the physically disabled, with users able to control a prosthetic device without requiring motor or nerve functionality. The live conversion of brain signals to useful prosthetic control requires a number of stages in a BCI. First, the device should collect neural data, and pre-process the inputted signals. Next, devices must extract feature information from signal sequences, retaining and transforming meaningful neural data, and classifying this into a movement instruction from

a predetermined set. Different feature extraction or classification approaches can be utilised, but usually require a degree of training or validation prior to use. Finally, the device must pass this instruction to the external device to enable movement [1, 10].

Since Jacques Vidal’s groundbreaking *Brain-Computer Interface project* coined the term in the 1970s, ongoing research into BCIs has led to many significant and valuable applications [11]. The potential of these devices to enhance societal inclusiveness has become increasingly evident as the implementation of BCI communication continues to increase [12]. Alternative communication pathways are already assisting disabled individuals through providing alternative control methods, enabling assisted speech, improved movement, and more [12, 13, 14]. *BrainGate* is a notable example of a BCI system used in accessibility. This system, initially researched by Brown University, assists patients with tetraplegia in communicating with a large range of external devices using only brain signals [15]. Research into this product helped pioneer the demonstration of feasible direct brain control over external assistive devices, as *BrainGate* devices have facilitated neural control over both prosthetic arms and cursors [16].

1.1.2 ERP & Oscillatory BCIs

BCIs can be classified into two main categories: Event-Related Potential (ERP) BCIs and Oscillatory BCIs, differentiated by characteristics of the neural signals they detect and process.

ERP events are the predictable fluctuations in neural impulses which occur at the onset of external or cognitive stimulation [17]. ERP BCIs focus on the detection of these types of signal events: inputs which yield uncharacteristically high amplitudes and low frequencies. The use of BCI models of this form involves classification of predictably shaped wave-forms, and generally performs with high robustness in cross-subject evaluation [1, 17]. Signals processed by this form of BCI exhibit comparatively high Signal-to-Noise Ratios (SNRs) when compared to those in Oscillatory-based BCIs. Detected signal ‘events’ in ERP systems can be identified with more confidence as a result, as the differences in amplitude are more clear [17].

Oscillatory BCIs instead detect the relative signal powers of input frequencies, like changes in μ (8-13Hz) or β (13-30Hz) frequency band powers [18]. This BCI

paradigm permits asynchronous control, as the system can continuously read and respond to brain activity in a self-paced manner without requiring a time-locked stimulus [18, 19]. For this reason, correctly trained and calibrated oscillatory BCIs can provide more utility for disabled user control, as they do not require an input cue. Operating an oscillatory BCI also requires far less concentration from the user, and could ideally be controlled during everyday tasks on demand [20]. As such, this research will focus on designing a robust classification model optimised for oscillatory BCIs, enabling reliable and accessible device control for users without reliance on external stimuli.

1.1.3 Neural Input Paradigms & EEG

BCIs can be set up to record neural electrical input from a brain with varying degrees of patient invasiveness. These approaches differ based on their recording methodologies, as well as in their safety, practicality, and typical applications. Most BCIs can be categorised as either *invasive* or *non-invasive* BCIs [21, 22].

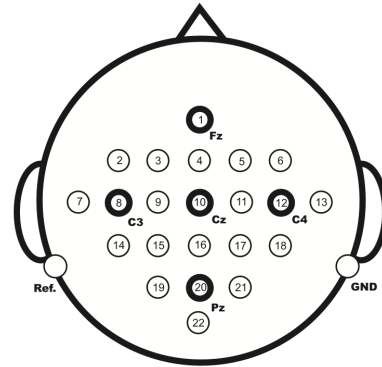
Invasive BCIs involve the surgical implantation of micro-electrodes on specific areas of a brain cortex. These devices can record neuron activity at a high spatial and temporal resolution through signals including Action Potentials and Local Field Potentials [22]. Invasive BCIs offer a comparatively high SNR, and rich extracted information content due to their close recording of neuron activity [23]. These devices have exhibited the ability to facilitate precise control to users in a range of applications, including control of robotic limbs [23, 24]. However, there are a number of clear drawbacks to the use of invasive BCIs. Cortical implants of these systems have damaged tissue during implantation and breached users' blood-brain barriers, causing degeneration to nearby neurons, as well as the failure of implants themselves [22]. Invasive BCI implantation also increases the risk of infection and nerve and blood cell damage to the user [25]. In addition, invasive BCIs face significant accessibility limitations, as their high cost, surgical complexity, and need for specialised clinical environments restricts their use to controlled research or medical settings rather than widespread personal application.

Non-invasive BCIs involve the collection of neural input signals without direct connection to the brain. Ascertaining data in this way also requires no surgery, as recording devices can be attached or removed without even piercing skin. As a result, non-invasive BCI input collection carries a comparatively low medical risk and greater accessibility to users [7, 22]. Non-invasive BCIs have already made

a considerable impact in a number of fields, particularly in rehabilitation from neurological and physical conditions [26]. However, these devices are troubled by signal attenuation through the skull and tissue, as these natural barriers between electrical impulses and electrodes act similarly to a low-pass filter, meaning certain high-frequency signals can be filtered out of the recorded data [22]. The indirect connection to neuron activity also means such input signals exhibit a low spatial resolution, with the activity of small neuronal clusters (detectable in invasive LFPs) often undetectable in non-invasive BCIs.



(a) Arrangement of electrodes on a patient's scalp during an EEG scan [27].



(b) Top-down diagram of an electrode configuration for EEG data collection. This 22 electrode configuration was used to collect the data for the BCI Competition IV Dataset 2a [2].

Figure 1.1

Electroencephalographs (EEGs) have become the most widely used non-invasive input format for BCIs due to their balance of affordability, convenience and temporal resolution [21]. These scans ascertain neuronal input signals by recording voltages from different positions on a user's head (as seen in Figure 1.1a) over a period of time [21]. Electrodes are strategically placed in measured locations around the head to facilitate this, in order to capture specific spatial activity. An example electrode configuration can be seen in Figure 1.1b. The quantity and arrangement of electrodes influences the resolution and total activity represented by the EEG, with higher density EEGs allowing for finer spatial resolution. Despite enabling greater information extraction, excessively high spatial resolution can dramatically increase processing complexity and data volume [21]. Recorded EEG trials take the form of multi-channel time-series data, where each channel corresponds to an electrode's voltage trace over time (see Figure 1.2).

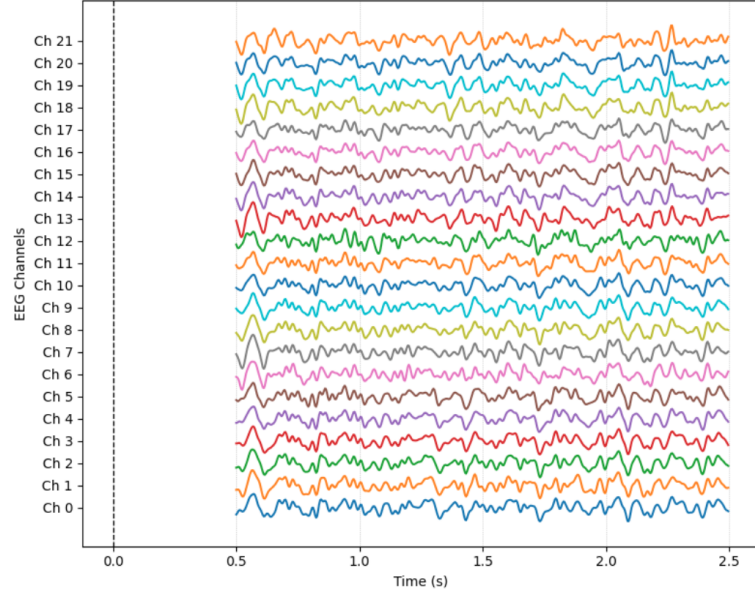


Figure 1.2: Visualisation of a single EEG trial. The sample comes from the BCI Competition IV Dataset 2a [2], and shows the neural activity from 0.5 to 2.5 seconds after the onset of an MI cue. Each waveform corresponds to the power detected over time by a single electrode.

1.1.4 Motor Imagery

Motor Imagery (MI) refers to the process of mental simulation of movement without physically performing the action. Oscillatory BCI systems have been utilised in the classification of EEG inputs to MI classes [8, 28, 1]. A common example is the administration of an EEG on a participant as they are asked to perform a certain motor imagery task, for example: ‘mentally simulate the action of extending your left leg’. The collection of signals produced and the associated MI class are recorded together, with the signal data used as an input, and the MI class used as a target output. This data can be used to train, validate and test an EEG-MI classification system, which can then be used to classify EEG input to MI output in real time [1]. When embedded in BCIs, outputs of this model could be transmitted to a movement device, with the device ultimately determining the move.

The most evident advantage of using motor imagery as an input method for prosthetic control is that it requires no muscle movement or peripheral nervous system activation. As a result, motor imagery is particularly well-suited for individuals with physical disabilities, including amputees and paraplegics, who often

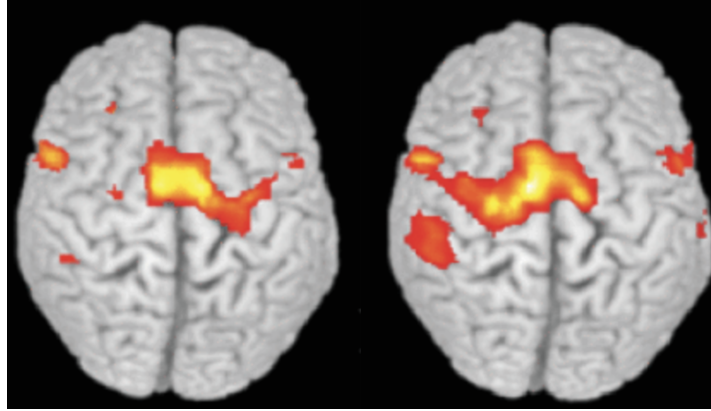


Figure 1.3: fMRI scans showing cortical activation during MI tasks. (left) Activation during left hand imagery, (right) activation during right hand imagery [34].

retain intact brain functionality despite having limited muscle or peripheral nervous system control [29, 30]. This is supported by studies into *Phantom Limb Syndrome*, a condition reported by over 80% of limb amputees in which patients experience pain, numbing and instinctual reflexes on limbs which have been amputated. Research has suggested that this condition is influenced by the brain maintaining almost entirely intact representations of limbs, even after their complete amputation, with perceived motor control and subsequently motor imagery abilities of these remaining active [30, 31]. This persistence of motor representations makes motor imagery an ideal candidate for translating these signals into prosthetic control.

Additionally, MI has been shown to exhibit cortical lateralisation, with imagery tasks of different body parts producing EEG patterns in different localised brain regions. Literature on motor imagery classification has shown that imagined movement on one side of the body produces distinct patterns in the opposite side's brain activity. Specifically, motor imagery induces Event-Related Desynchronisation (ERD) in the α frequency bands and Event-Related Synchronisation (ERS) in the β frequencies in the opposing hemisphere of the brain [32]. An example of this is illustrated in Figure 1.3, which shows the distinct regions of brain activation during left and right hand motor imagery tasks. Such distinct and localised EEG patterns corresponding to different motor imagery tasks highlight the feasibility of accurately classifying MI types, reinforcing its suitability as an input method for EEG-based prosthetic control systems in BCIs [33, 10].

1.1.5 Deep Learning with BCIs

A focal challenge with MI-based BCIs lies not just in acquiring signals but also in effectively interpreting them. EEG signals in particular are inherently noisy, non-stationary, and complex, making the task of extracting discriminative and useful features particularly difficult [35]. Traditional approaches often rely on recognition and extraction of handcrafted features hard-coded into models, which struggle to generalise across subjects and domains [1]. To combat this, deep learning methods have become increasingly popular in the EEG-MI classification field, with these models exhibiting the impressive ability to automatically discover useful representations in EEG data [36, 37]. **convolutional neural networks** (CNNs) have shown particular promise in this area due to their capacity to capture both spatial and temporal patterns in structured data [1, 38].

CNNs are deep learning architectures which gained prominence for their impressive performance in classifying images and grid-like data, while avoiding computational and overfitting limitations common in other network types in the same contexts [39]. CNNs harness filters which slide across input images, performing a series of matrix convolutions to detect collections of spatial patterns, such as edges, textures and corners. These intermediate outputs are referred to as **feature maps**. Layered convolutions and feature extraction are interleaved with activation functions to introduce non-linearity and pooling layers to reduce dimensionality, enabling the network to learn increasingly abstract representations while avoiding storing an excessive volume of parameters [8, 1, 40].

The advantages of using CNNs for image recognition are highly transferable to EEG signal classification tasks, and as a result make CNNs a suitable model choice for EEG-MI classification [1, 41]. Time series data recorded over T time points from C EEG channels can be structured as a matrix of shape $C \times T$, which can be treated analogously to a grayscale image of the same dimensions. CNNs excel at extracting spatial and temporal features simultaneously from these representations, using vertical filters to capture spatial relationships across channels and horizontal filters to capture temporal patterns. Through doing this, CNNs can learn discriminative patterns associated with different MI classes [42].

Another benefit of using CNNs for this task is their capacity for automatic feature extraction. CNNs extract features at multiple levels of abstraction, from low level patterns to high level characteristics, operating independently of the problem domain or input format. This replaces the need for domain expertise and manual feature engineering, as these networks are capable of capturing a

wide variety of features with no prior subject knowledge encoded [1, 40]. However, these systems often lack innate feature explainability, a common pitfall of deep learning systems. The networks encode informational patterns with complex and abstract feature maps, which are difficult to interpret natively, particularly over a complex domain like the EEG time series [40]. A lack of feature explainability could result in reduced trust of network outputs, challenges in error diagnoses and problems meeting legal and ethical biomedical standards. When implemented in situations as sensitive as controlling a human body from a brain-embedded device, these issues could pose significant implications for acceptance and adoption of these technologies [43].

1.2 Existing approaches to EEG-MI classification

Prior to recent breakthroughs in deep learning approaches, most EEG-based classification architectures were designed with a particular input focus, tailored to specific BCI paradigms or input signal types. The clear limitation of these types of models is their lack of generalisability and transferability, tending to only perform optimally with a specific input paradigm [44, 45]. This restricts their applicability, often requiring the frequent development of new classification models tailored to the hardware and signal characteristics of each individual BCI system. For instance, model architectures with feature extraction and pre-processing methods designed specifically for ERP input are known to significantly underperform when applied to oscillatory EEG signals. ERP-based models frequently harness signal averaging, which can suppress oscillatory activity, leading to the loss of vital input features and significantly reduce classification accuracy [44]. Such designs also typically require a large amount of input-specific prior knowledge, with both signal pre-processing and feature extraction methods heavily custom-tailored using this to facilitate a specific input method. This can prove inconvenient and infeasible to collect, requiring additional tests and training data to optimise model components to specific input paradigms [46, 47].

1.2.1 Deep & Shallow ConvNets

In 2017, a pioneering deep learning solution to EEG-MI classification was proposed by Schirrneister et. al [46]. The authors look to avoid the pitfalls of input-specific models through harnessing a CNN to avoid hard-coded feature extraction, all while maintaining a strong classification accuracy. It featured two convolutional network architectures: **Deep ConvNet** and **Shallow ConvNet**. These networks were designed with the aim of developing generic CNNs with only

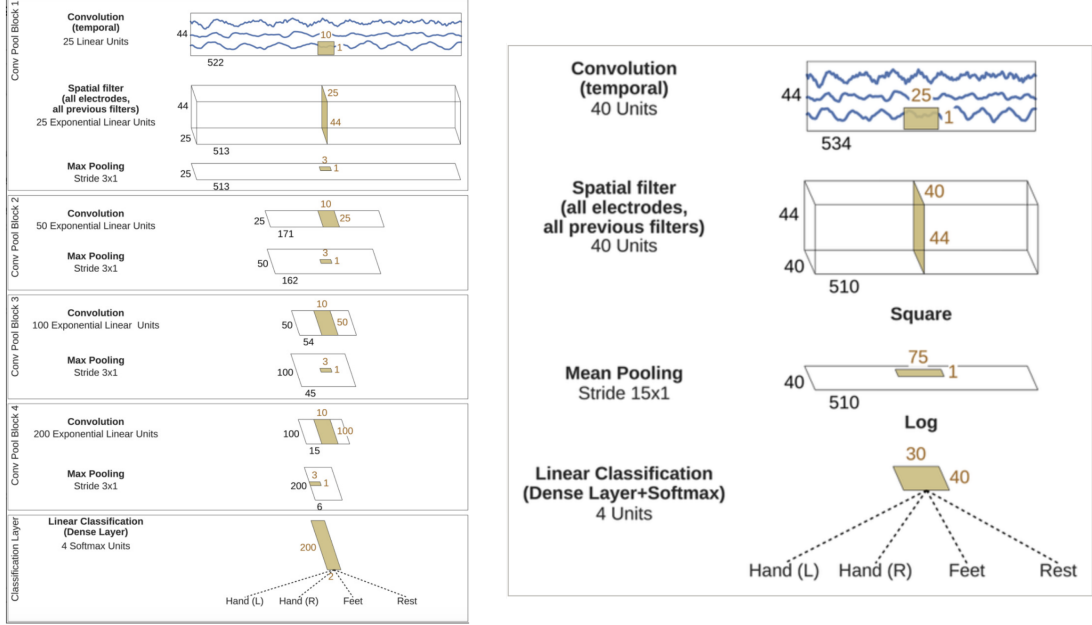


Figure 1.4: (Left) Deep ConvNet architecture, (Right) Shallow ConvNet architecture [46]

minor levels of subject expertise that could still achieve relatively high accuracies in EEG classification tasks. The architectures of both ConvNets are visualised in Figure 1.4.

Deep ConvNet consists of four convolutional blocks, each consisting of sequential convolution and max pooling, followed by a fully connected classifier with a softmax activation. Its first block is designed for EEG-style input: first applying a temporal convolution across each channel’s time-series, and next applying a spatial filter across the electrodes. The remaining three blocks are more standard convolution–pooling layers, progressively transforming EEG signals into higher-level representations by using increasing numbers of activation units.

Shallow ConvNet features a smaller network, harnessing a shallow design inspired by the Filter Bank Common Spatial Patterns (FBCSP) pipeline [48]. It features only one convolutional block, applying temporal and spatial convolutions sequentially. However, it features a temporal kernel of length 25 (rather than 10 in Deep ConvNet) in order to better capture frequency-specific features in the first convolution. Shallow ConvNet also adds more non-linearity to the output through squaring and logarithmic activation functions. Following this, mean pooling is applied and the output is fed through a fully connected layer with a softmax activation function for class prediction.

Both of these models help inspire the proposed model, not only directly but indirectly through their contributions to the development of other models [1]. Both models also serve as benchmarks for evaluating the relative performance of the proposed architecture.

1.2.2 EEGNet

EEGNet is another classification model designed to classify motor imagery outputs across various BCI input paradigms. These include both ERP and oscillatory based inputs, such as visual-evoked potentials, error-related negativity, movement-related cortical potentials, and sensorimotor rhythms [49, 1]. The introduced model of EEGNet is inspired by CNNs in Shallow and Deep ConvNets, while aiming to improve performance and paradigm generalisability. The network’s design comprises of two feature extraction blocks with a combination of depthwise and separable convolutions, followed by a fully connected classification head.

The first extraction block takes EEG input in the form of C channels over T samples, and applies two convolutional steps. The first step being the convolution of signals with F_1 1D temporal convolutional filters of length 64, and the second being convolution with D depthwise filters. This concatenation of temporal and depthwise convolutions facilitates the extraction of frequency-specific spatial features [1]. This block first uses linear activations after each convolution, followed by an ELU activation to establish non-linearity. It also harnesses dropout and batch normalisation to increase model regularisation, and a temporal pooling layer of size (1,4) to reduce size of outputs.

The second block uses *separable convolution*, a technique harnessing sequential depthwise convolution (filtering of each input channel independently) followed by pointwise convolution (combining the filtered outputs across channels) for parameter reduction and decreased computational cost. This layer not only reduces the output size, but also sequentially ‘summarises’ extracted feature maps, and combines their outputs.

Finally, the model flattens the data into a 1D vector, and uses a fully connected layer with a softmax activation to classify feature outputs to MI instructions, with the number of output neurons set to the number of classes in the input data. A high-level overview of the model is visualised in Figure 1.5.

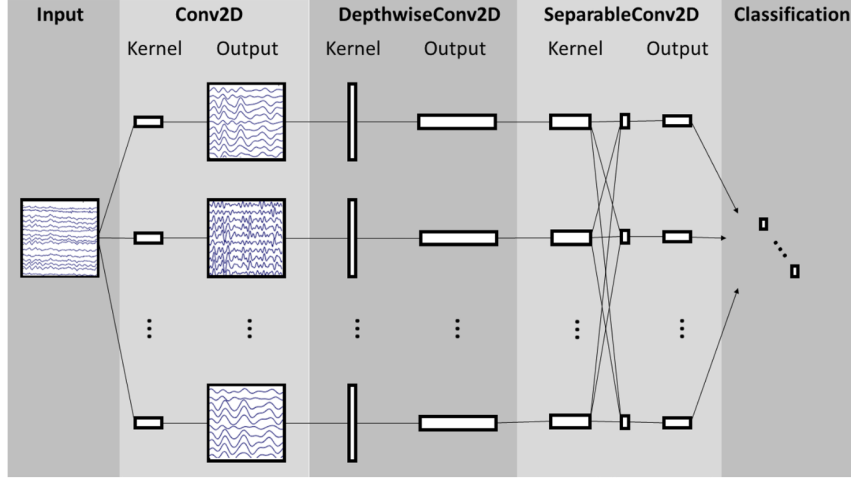


Figure 1.5: The (simplified) EEGNet Network Architecture [1]

The model is proposed with two structural variants: *EEGNet-4,2*, where $F_1 = 4$ and $D = 2$, and *EEGNet-8,2*, where $F_1 = 8$ and $D = 2$, with EEGNet-8,2 featuring approximately double the number of parameters. The variants did not exhibit statistically significant difference in performance measurements, indicating that the addition of more filters (and increase of model complexity) does not inherently improve CNN model performances [1]. The EEGNet model significantly outperformed ERP-BCI native *xDAWN+RG* model on all same-subject ERP inputs, and all but one cross-subject ERP input. The model also significantly outperforms the Oscillatory-BCI native *FBCSP* algorithm in cross-subject oscillatory input, and performs similarly over same-subject oscillatory input. This shows the ability of the EEGNet structure to not only effectively generalise over a range of input paradigms, but also perform as well as or better in these paradigms than specialist models for each input format. In addition to this, the EEGNet model features a vastly reduced number of parameters when compared to these comparison models, with both EEGNet variants featuring less than 2.5% of the parameters of each of the other compared models.

EEGNet’s ability to outperform specialised models with drastically fewer parameters demonstrates its effectiveness as a lightweight and versatile solution for EEG-MI classification over diverse input paradigms. This encouraging balance between performance and cross-domain efficiency makes EEGNet a strong foundation for further experimentation. Its compact architecture reduces the risk of overfitting on limited EEG datasets, which is a critical concern in this research. **For these reasons, the EEGNet model structure is harnessed as the foundation for the model developed in this work.**

1.2.3 CTNet

CTNet is a recent EEG-MI classification model that combines the paradigms of CNNs and transformers to form a hybrid feature extraction framework [1, 50]. Leveraging the established EEGNet architecture, CTNet retains its convolutional front-end to extract spatial and temporal features, while adding a transformer encoder module to capture longer-range global dependencies across EEG channels and time. This integration enriches contextual representation and feature discrimination within EEG data.

In evaluations across two datasets, CTNet achieved higher classification accuracy than EEGNet for the majority of subjects, demonstrating the advantage of augmenting CNN-based architectures with attention mechanisms. Although not directly implemented in this work, CTNet demonstrates that refining and extending the established EEGNet architecture can mean meaningful improvements in classification accuracy.

1.3 Datasets

1.3.1 Data Preparation

The classification of EEG signals presents several significant challenges. Firstly, EEG data typically exhibits low SNRs, which complicates the task of extracting meaningful features from the raw signals due to the distracting presence of ‘noisy’ signals. [41]. Additionally, there is high inter-subject variability in EEG recordings, stemming from both differences in individual brain activity and variations in recording setups [36, 8]. EEG signals often feature the presence of artefacts, which constitute momentary distortions in signal data. These artefacts, often caused by small muscle movements including blinks or twitches, can prompt incorrect feature recognition and reduced classification accuracy [1, 51, 52]. To mitigate the effects of these challenges, applying data pre-processing techniques to inputted signal data has proven critical in ensuring high accuracy predictions can be made [1, 46].

- **Band-pass filtering** has been widely used to remove irrelevant and noisy frequencies [50, 53]. This technique aims to include only frequency bands relevant to motor imagery, such as μ and β rhythms, known to exhibit ERD during imagined movement [33, 10]. A major drawback of this technique is

its requirement for problem-specific knowledge and task adaptation, limiting its generalisability in models intended for cross-domain usage.

- **Artefact Removal** can be employed to identify and remove signal distortions caused by irrelevant stimuli. Independent Component Analysis (ICA) is one removal technique, which decomposes signals into a specified number of components. This process can be employed to detect components created by artefacts, subtract these from the overall signal, and reconstruct the ‘cleaned’ EEG signal in its original format with reduced artefact impact. [54].
- **Baseline correction** is also often used in EEG pre-processing, assisting in tweaking individual trial variability. This technique assesses differences between EEG signals before and after patient stimulation, reducing the impact of irrelevant signal fluctuations [55].
- **Amplitude Magnification** involves multiplying the amplitudes of all signals by a constant factor. Although this does not alter the SNR of input data, it can help bring weak EEG signals into measurable and numerically stable ranges for deep learning models [56]. Magnification also helps to preserve relevant fluctuations and ensures they can be learned by weighted convolutional filters [56].
- **Data Augmentation** is a common method of preparation in other fields, an approach in which new training samples are generated to artificially increase the training set size [57]. These samples can be generated from applying techniques like segmentation and recombination, truncation and addition of white noise to copies of existing training instance to generate a new sample. Such augmentation strategies have proven highly effective in mitigating overfitting and enhancing the generalisation of machine learning models [50, 57].
- **Class Stratification** is a technique used to ensure the proportion of samples belonging to each target classes in each dataset partition is equal to its proportion in the whole set. This ensures that each class is appropriately represented in training, validation and test sets, as well as across cross-validation folds, preventing models from being biased toward selecting majority classes [58].

A combination of these data pre-processing and preparation techniques will be utilised by the proposed model. The specific implementation of these is discussed in more detail in Section 2.2.

1.3.2 Available EEG-MI Data

A number of publicly accessible datasets containing EEG data and corresponding MI inputs exist, many of which are published for free use online [59, 3, 2, 46, 60]. These datasets use a range of formats to depict recordings of EEG signals with associated target MI classes from a specified list. The MI classes reference the MI instruction provided to a patient during a sample, and specify the point during the sample this instruction occurred (if applicable). These datasets play an important role in BCI development by facilitating model comparisons and establishing measurable model performance criteria. A brief comparison of a number of notable available EEG datasets is available in Table 1.1.

Dataset	Subjects	EEG Channels	Target MI Classes
BCI Competition IV Dataset 1 [59]	7	59	3: Left Hand, Right Hand, Both Feet
BCI Competition IV Dataset 2a [3]	9	22	4: Left Hand, Right Hand, Both Feet, Tongue
BCI Competition IV Dataset 2b [2]	9	3	2: Left Hand, Right Hand
High-Gamma Dataset (HGD) [46]	14	44	3: Left Hand, Right Hand, Both Feet
BNCI Horizon 2020 ‘002-2014’ Dataset [60]	14	15	Left Hand, Right Hand, Both Feet, Tongue

Table 1.1: A selection of notable publicly available EEG motor imagery datasets

1.3.3 Fixed-Length vs. Continuous Datasets

Many evaluations of EEG data for use in BCI-based classification evaluate windowed fixed-length EEG input. EEG datasets with this format are referred to as *Fixed-Length Datasets*, and involve individual EEG trials of fixed time periods, recorded during a period of MI in a subject. BCI Competition IV’s Datasets 2a and 2b, discussed in more detail in section 2.1, are two widely evaluated fixed-length datasets. In such sets, each trial is represented by a time series with a known length and sample rate, and labelled with the target class of the shown prompt. Fixed-length datasets remain as the most commonly used format in BCI research due to their clearly defined trial boundaries, which simplify pre-processing and allow models to be trained on consistent temporal segments.

This is particularly advantageous for CNN-based models, as it ensures inputs of uniform size, which can be effectively processed through convolutional and pooling layers without additional alignment or segmentation steps.

Continuous Datasets use a continuous format, with each trial representing a longer time series at a known sample rate with a number of labelled (but uncued) actions. Rather than feature separate windowed trials with each trial attributed to one prompt, multiple different imagery prompts can be shown throughout the same trial, with periods of rest between each MI period [61]. No cue information is provided to the classifier, and the model has no inherent indication of the timings or number of MI prompts shown in the trial. Therefore, this dataset requires classifiers to continuously analyse incoming EEG data without explicit cueing of the commencement or completion of a motor imagery action [61]. An example of a continuous dataset is BCI Competition IV’s *Dataset 1*, which provides trials in continuous streams of EEG with targets represented as pairs of timestamps and MI classes, indicating moments where specific MI prompts were shown [59]. Training BCIs with continuous data presents additional difficulties. In the absence of clear trial boundaries and cue information, models using such data often struggle to ensure accurate EEG classification. Models face the challenge of properly addressing periods of inaction between prompts, which mustn’t be mistakenly recognised as MI action. Recent techniques, such as the introduction of overlapping sliding windows, have attempted to address this by capturing transient MI events and classifying signal segments individually [62].

A key design decision in developing the proposed model was the choice between fixed-length and continuous dataset formats, each requiring distinct architectures. Following consideration, the model was designed for fixed-length input trials to ensure consistent training and evaluation while avoiding challenges inherent to continuous data. Fixed-length datasets are also more standard in benchmark EEG-MI studies [1, 50, 46], enabling direct comparison with established baseline models.

1.4 Research Overview

1.4.1 Hypothesis

Before undertaking this research, the following **hypothesis** is proposed:

Targeted enhancements to the pre-processing pipeline and network architecture of EEGNet will lead to measurable improvements in motor imagery classification accuracy on at least one benchmark dataset, while maintaining a generalisable model structure.

If the research fails to support the hypothesis, it may suggest that EEGNet already represents a near-optimal solution for EEG-based motor imagery classification. It may also suggest that seeking generalisability restricts the model's representational capacity, implying that meaningful performance gains might only be achievable through more specialised, less general architectures.

1.4.2 Objectives

Despite steady advancements in EEG-based MI classification, existing models still exhibit accuracies that are too low and inconsistent for practical BCI applications. These limitations restrict their reliability in real-time control tasks, particularly in contexts requiring robustness, minimal calibration, and generalisability.

The objective of this research is to bridge the gap between current classification performance and that required for real-world application by developing a compact, high-performing EEG-MI classifier. Through targeted improvements to both the pre-processing pipeline and network architecture of an established model, this research seeks to enhance its classification accuracy and stability without significantly increasing its complexity.

1.4.3 Aims

This project attempts to achieve this objective through meeting the following aims:

1. Apply and evaluate **pre-processing techniques** to prepare signals to minimise classification challenges in EEG data, including low SNR, inter-subject variability and ocular artefacts.

2. Develop and train an end-to-end **EEGNet-based CNN model** to classify signals into their corresponding MI class.
3. Investigate the impact of model design and architecture on model performance to **improve performance** while retaining a lightweight and generalisable implementation.
4. **Benchmark** the performance of the model against established and comparable baselines (such as **EEGNet** and **ConvNet** variants) to quantify improvements.

CHAPTER 2

Methods

2.1 Data

Two fixed-length datasets were eventually selected for training and performance benchmarking of the proposed model: **Dataset 2a** [2] and **Dataset 2b** [3] from the BCI Competition IV held in 2008. These datasets, collected in collaboration between the Institute for Knowledge Discovery and the Institute for Human-Computer Interfaces at Graz University of Technology [63], are among the most widely used in benchmarking of EEG-MI classification models [1, 50, 46]. Both of these datasets are publicly available, distributed under a Creative Commons Attribution-No Derivatives 4.0 International License.

2.1.1 Dataset 2a

Dataset 2a was collected from an experiment conducted on nine subjects, with four possible motor imagery cues: imagined movement of the left hand, right hand, feet and tongue. Each subject was recorded for 6 sessions on one day, and 6 sessions on another day, with each session featuring 6 runs of 48 trials each. These 48 trials are evenly split between the four MI classes, resulting in the overall dataset having an even ratio of output classes and no class imbalance. The 288 trials from the first day form the training set for each subject, while the 288 trials from the second day form the evaluation set. In total, this dataset comprises of a training set and evaluation set of 2592 EEG trials each, with every trial labelled with the corresponding MI cue [2].

In the experiment, each trial begins with the display of a fixation cross on a screen in front of subjects for 2 seconds to indicate the beginning of a trial. Following this, a cue is shown with an indication of which of the four imagined movements to conduct. The subject conducts the instructed imagery for 4 seconds following the cue onset. A 1.5 second break follows, before the process is

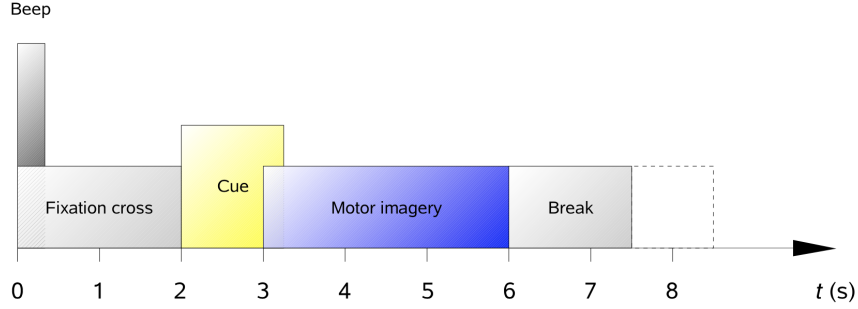


Figure 2.1: Method of data collection of each trial in Dataset 2a [2].

repeated for another trial. The full experimental timing paradigm is visualised in Figure 2.1. During this process, EEG data is collected through 22 electrodes placed 3.5cm apart on the subject’s scalp, arrangements of which can be seen in Figure 1.1b. Additional electrooculogram (EOG) data is recorded from three electrodes around the subject’s eyes. All signals are recorded at a sampling rate of 250Hz, and have a bandpass filter of 0.5-100Hz and a notch filter of 50Hz sequentially applied in order to suppress unwanted line noise [2].

2.1.2 Dataset 2b

Although collected under many of the same conditions by the same researchers, there exist several key differences between datasets 2a and 2b. Dataset 2b features 9 subjects, but only 2 motor imagery classes: left and right hand imagined movement. Each subject was recorded over three separate days, with each of the first two days featuring 120 *offline* trials per subject. These collection of these trials followed a similar paradigm to Dataset 2a, with each iteration featuring a fixation cross, MI cue, 4 second imagery period, and brief rest. These timings are displayed in Figure 2.2. However, this dataset’s collection protocol differs with the addition of *online feedback sessions* during the third recording day. In these sessions, a classifier trained on data from the first two days was used to predict each subject’s motor imagery in real time. During the motor imagery period a dynamic feedback signal was shown: a green smiling face if the classifier’s prediction matched the cue, or a red frowning face if it did not. This feedback was intended to provide immediate reinforcement, allowing subjects to adjust their mental strategies and enhancing the consistency and quality of their motor imagery signals produced. Online trial timings (also visualised in Figure 2.2) are largely unchanged, with a neutral grey face replacing the fixation cross. The face then displays dynamic feedback for the final 3.5 seconds of the MI period.

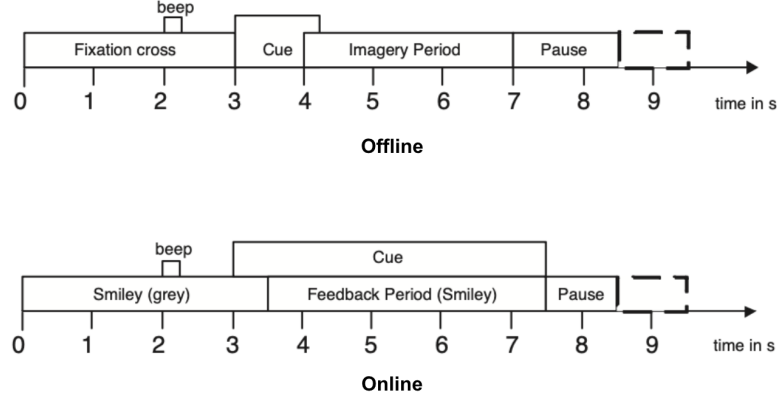


Figure 2.2: Timing paradigms of data collection in Dataset 2b [3]

On the third day of recording, three online sessions are conducted for each subject, each comprising 80 trials. The first of these sessions is combined with the first two days’ data to form the training set, with a total of 320 trials per subject. The remaining two online sessions form the evaluation data, with 160 trials per subject in this set. In this experiment, all runs are divided evenly between the two imagery classes, so there is similarly no class imbalance [3]. The EEG recording setup in this experiment differs from that in Dataset 2a, with only three electrodes used to record EEG data. These electrodes are located at the positions marked by **C3**, **Cz** and **C4** in Figure 1.1b. Three additional electrodes for EOG collection are also used. Recorded signals are sampled at 250Hz and passed through the same bandpass and notch filters as Dataset 2a.

Although Dataset 2b remains a valuable resource, greater emphasis was placed on optimising performance on Dataset 2a. This decision reflects its wider use as a benchmark in the relevant literature, with the baseline EEGNet notably evaluating on Dataset 2a but not 2b. In addition, Dataset 2a provides four motor imagery classes instead of two, offering a more challenging classification task. A model trained on a more complex multi-class task is preferable, as the ability to discriminate among a larger set of motor imagery actions facilitates broader generalisability and better transfer to practical BCI applications.

2.2 Pre-processing

As part of the proposed model, a data processing pipeline is implemented which modifies inputted data **prior** to its forward pass through the convolutional model. This pipeline includes several techniques adopted from the pipeline of the EEG-Net model that inspired this work. Additionally, several novel techniques are introduced with the goal of improving classification accuracy and generalisability.

2.2.1 Adopted Techniques

First, the components of the pipeline adopted from EEGNet are outlined.

Inputted raw signals are first epoched from the dataset into event-locked trials. This action involves the timewise selection of only signals within a range defined by $[\mathbf{tmin}, \mathbf{tmax}]$. The choice of $\mathbf{tmin}$ and $\mathbf{tmax}$ was individually tailored to each dataset to ensure maximum retention of signal segments containing discriminative motor-imagery activity, while excluding segments prone to noise or irrelevant fluctuations that might degrade classifier performance. Baseline correction was initially applied during this epoching but ultimately abandoned, as it produced negligible improvements in classification performance. These results were unsurprising, as baseline correction has often proven ineffective for EEG pre-processing, as its drifts are better addressed by filtering or normalisation methods [64].

In Dataset 2a, signals were epoched between the range of 0.5 and 2.5 seconds following cue onset. This period was found to contain the most discriminative EEG activity. Signals recorded before 0.5 seconds contained minimal discriminative data, likely a result of the reaction time of the subject to the cue. Inclusion of EEG data from after 2.5 seconds was also non-discriminative and even found to decrease accuracy in many trials. This is due to the reduced significance of ERD and ERS over time as a result of concentration drifts and subject fatigue, leading to weaker MI responses and a more challenging classification task [65]. This was also the time window used for epoching in both the ConvNet and EEG-Net implementations, further supporting its suitability. [1, 46]. For Dataset 2b, a time window of 0 to 4 seconds was chosen. This interval encompasses the entire imagery phase, beginning at cue onset and ending at trial completion. This choice was based on controlled experiments varying the minimum and maximum bounds, which were necessary due to limited prior evidence for this dataset. The results showed that this broader window achieved the highest classification accu-

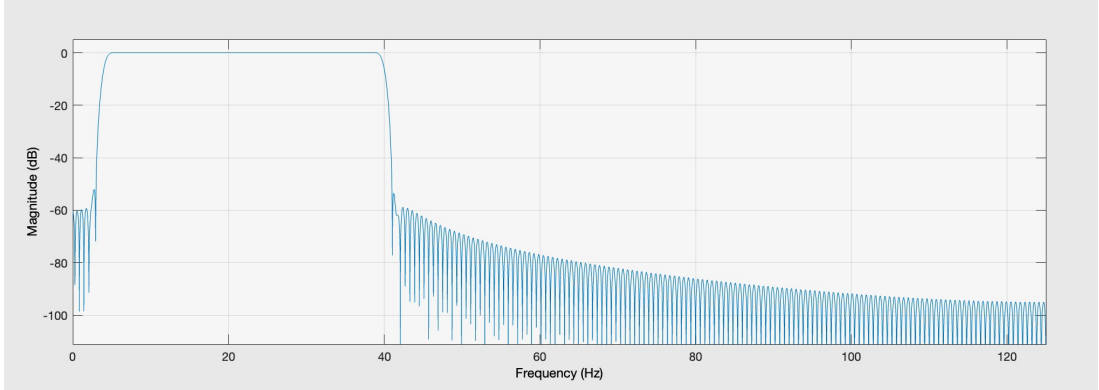


Figure 2.3: The 4-40 Hz bandpass filter applied during processing

racy. These results contrast with those observed for Dataset 2a, likely due to the nature of Dataset 2b’s online feedback trials, which encouraged greater subject engagement and sustained focus. As a result, the effect of fatigue or concentration drift on the quality of the EEG signals was reduced.

After epoching, a selection of signal processing techniques are applied to the EEG signals. The first of these is a bandpass filter with a range of 4-40Hz, resembling that used in the pre-processing of EEGNet [1]. Evidence suggests that MI-related information is almost entirely concentrated within or just outside the μ (8-13Hz) and β (13-30Hz) rhythms, with meaningful spectral power rarely extending much further than these bands [65]. Accordingly, signals above 40 Hz are filtered out, as higher frequencies contribute little discriminative value while introducing additional noise [65, 1]. Frequencies below 4Hz are also filtered out, with these frequencies also heavily contaminated by ocular artefacts such as blinks and eye fluctuations [66]. This filter is applied using a Hamming windowed FIR filter, visualised in Figure 2.3.

Next, signals are normalised using the same technique employed in both ConvNets and EEGNet: an exponential moving standardisation (EMS) is applied to all inputted EEG data with a decay factor of 0.999 [49, 46]. EMS is a form of normalisation which updates the mean and variance of each electrode’s time-series using a weighted combination of past and current signals. This method ensures the convolutional model receives inputs with stable amplitudes and variances over time, and reduces the impacts of slow drifts on recordings. EMS was chosen over conventional standardisation methods due to its ability to adapt dynamically to rapidly changing signal input statistics [46].

Additionally, EOG channels are omitted before input into the convolutional model, a step recommended by the creators of the datasets and implemented in all researched peer pipelines [2, 3]. These signals primarily capture eye movements rather than cortical activity, introducing non-neural artefacts into the input. Not removing these channels risks decreasing model performance by contaminating motor imagery signals with irrelevant ocular activity [67].

2.2.2 Novelty Implemented Techniques

In addition to the listed steps, this research extends the pipeline by introducing several novel techniques. Some of these techniques are implemented to ensure optimal model function and reliable learning behaviour. First, the amplitudes at all timesteps are **magnified** by a factor of 1000, helping to ensure numerical representations of EEG activity are effectively captured by convolutional filters. Preservation of class balance is also critical, ensuring the model does not develop biases towards majority classes. Although class weighting is not required (due to the equal counts of trials of each MI class in every set), **class stratification** is implemented when sets are divided. This includes the splitting of training sets to form validation sets, and the creation of cross validation folds.

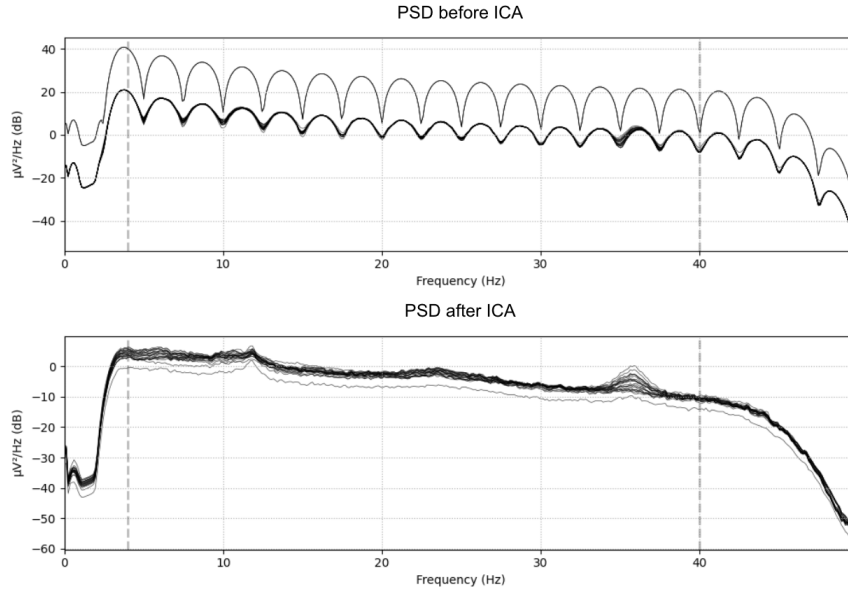


Figure 2.4: Power Spectral Density plot of an example trial's EEG signals before (above) and after (below) applying ICA.

Independent Component Analysis (ICA) is implemented next as a means of artefact removal, with the aim of removing distortion caused by interfering ocular activity. This begins with the decomposition of EEG signals into 20 components with maximum statistical independence. Next, components that are strongly correlated with eye movement activity (such as blinks) are identified and removed, as this movement is not related to motor imagery and irrelevant to classification. This usually involves the removal of between one and three ICA components from the signal. Finally, the cleaned components are reconstructed into their original channel-wise format, producing the processed EEG signal. As shown by Figure 2.4, the relative power of frequencies is changed significantly following the application of ICA. The power spectral density before ICA displays regular spectral ripples of inflated power, reflecting the dominance of high-amplitude ocular artefacts which introduce transients across frequencies. After ICA is applied, these components are removed, resulting in a smoother spectrum in which meaningful EEG rhythms are more clearly represented, and spectral ripples are removed. This demonstrates the ability of ICA to effectively suppress ocular interference by reducing spectral distortion.

Following the above pre-processing steps, **data augmentation** is harnessed to artificially increase the training set size and reduce overfitting. Augmentation begins with segmentation and recombination: each training trial is divided into five segments, after which segments from trials of the same class are randomly selected and concatenated in their original temporal order to create new synthetic trials. Following this, each segment has the possibility of undergoing any (or multiple) of the following perturbations: time shifting, Gaussian noise addition or channel dropout. Time shifting randomly displaces the selected segment by up to 20 samples (0.08 seconds) in either direction, while Gaussian noise addition generates normally distributed low amplitude fluctuations and adds them to the signal. Finally, channel dropout randomly selects and re-

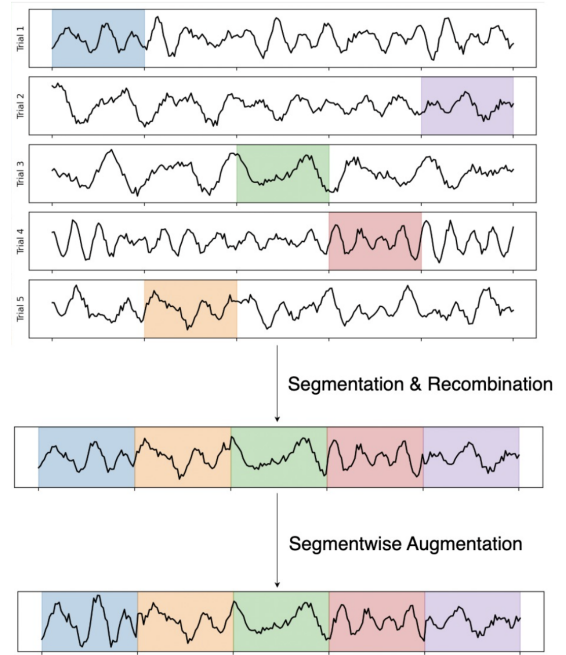


Figure 2.5: Example augmentation process of one training set trial.

moves one EEG channel from the segment. These perturbations have the probabilities of 0.5, 0.5 and 0.3 respectively to be applied on any given segment. A visual example of this whole procedure is provided in Figure 2.5.

Although having negligible impact on classification accuracy, the addition of augmented trials to the training set significantly reduced overfitting, training time, and the average number of epochs to reach convergence. This shows how the augmented data acts as a form of regularisation, increasing the variability of the training data. This led to more generalised representations, resulting in a trained model with greater robustness to fluctuations in input data.

2.2.3 Data Transformation

During the exploration of data preparation methods, the transformation of time-series signals into the time-frequency domain was investigated. This experiment aimed to determine whether time-frequency inputs enabled more effective feature extraction by the convolutional model. Both the Short-Time Fourier Transform (STFT) and the Wavelet Transform have been used with promising results in related EEG applications, such as seizure detection [68]. STFT divides a time-series into time windows of specified size and applies a fast Fourier transform to each [68]. The result is a three-dimensional representation of a signal (across the time, frequency and power domains), encoding both spectral and temporal resolutions. Rather than using the sine wave decomposition of a Fourier transform, the wavelet transform breaks a signal into smaller pieces using short wave segments or ‘wavelets’. Unlike STFT, the wavelet transform can adapt its analysis window for flexible characterisation of transient signal features [69]. Both of these methods were investigated in this research, with a visualised transformation of each shown in Figure 2.6.

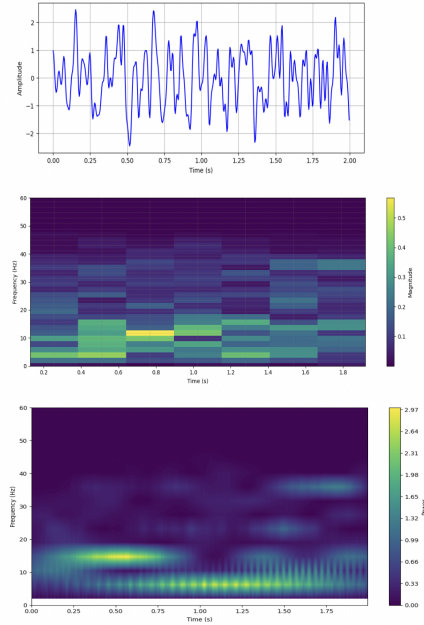


Figure 2.6: Illustration of EEG data from a single channel of a training trial across three representations: (Top) Time-series Signal, (Middle) STFT-transformed spectrogram, (Bottom) Wavelet-transformed scalogram.

Converting data to a time-frequency format using each of these transformations dramatically reduced the classification accuracy of the convolutional model. This result can be attributed to the intentional design of the EEGNet architecture upon which the proposed model is based. The temporal and depthwise convolutions in the network are tailored to capture distributed features across a time-series [1], and time-frequency input no longer aligns with this structure. This ultimately led to poor feature extraction and low classification accuracy. Despite significant modifications being applied to the architecture in an attempt to accommodate time-frequency input, the model continued to display poor classification accuracy. This prompted an eventual abandoning of these transformation methods and a return to the original time-series input format.

2.3 Convolutional Model

For feature extraction and classification a convolutional neural network was employed, drawing substantial inspiration from the established EEGNet architecture. Modifications and novel additions were incorporated with the objective of enhancing model generalisability and improving classification performance. Changes were also made to regularisation, optimisation and training of the model, to improve both accuracy and computational performance.

2.3.1 Network Architecture

The proposed model employs a CNN structure largely consistent with EEGNet’s (explained in detail in Section 1.2.2), building on its strong performance and generalisable design [1]. Some modifications are introduced to the network, extending the model’s capacity while maintaining a relatively lightweight design. A visualisation of this proposed architecture end-to-end is shown in Figure 2.7. The first two convolutional modules feature a near-identical network structure, also beginning with a sequence of temporal, depthwise and separable convolutional layers. These also feature the same combination of normalisation, pooling and dropout layers, as well as original activation functions. However, several layer hyperparameters, including convolutional kernel setups and dropout configurations, are adjusted to refine network behaviour and performance. The details of these modifications are explained in Section 2.3.2. The EEGNet classification head, which flattens the extracted features and uses an MLP with softmax activation

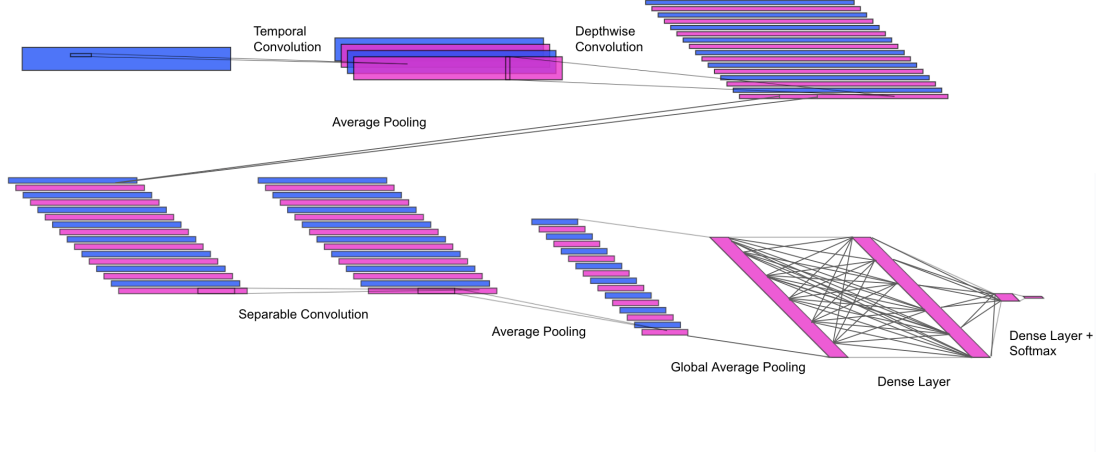


Figure 2.7: Visualisation of the proposed network architecture, demonstrating the process of MI classification from (pre-processed) EEG signals.

to output class membership probabilities, was also retained. A full comparative summary of the proposed network’s structure with that of EEGNet can be found in Appendix 1.

The most notable change to the architecture was the inclusion of an additional **third network block** following the second convolutional block. The module begins with a Global Average Pooling operation, which reduces the three-dimensional feature map to a one-dimensional vector of length 16. This pooling extracts only the most prominent features from the inputted maps, while also reducing the number of trainable parameters for the following layer. The vector is then processed by a dense layer containing 64 neurons, expanding representational power beyond purely convolutional feature extraction. This is followed by the application of a leaky ReLU activation function, establishing non-linearity while also mitigating the ‘dying ReLU’ problem [70]. To maintain consistency with the preceding convolutional blocks, batch normalisation and dropout are subsequently applied. These layers provide regularisation and improved generalisation, reducing overfitting and improving convergence speed. The resulting 64-dimensional output vector is passed onwards to the classification head. Overall, this additional block enhances the network’s ability to capture higher-level representations that extend beyond those extracted by the convolutional layers.

2.3.2 Regularisation & Optimisation

Several modifications are introduced to enhance the robustness of the convolutional layers, including changes to kernel initialisation, weight regularisation, and systematic hyperparameter optimisation.

Kernel initialisation refers to the process of assigning weights to convolutional filters before training begins. EEGNet, originally built with the Keras API of Tensorflow [1] simply uses the default Keras convolutional kernel initialiser: *Glorot Uniform* Initialisation [71], initialising weights following the distribution:

$$W \sim U\left[-\sqrt{\frac{6}{n_{in}+n_{out}}}, \sqrt{\frac{6}{n_{in}+n_{out}}}\right]$$

where n_{in} and n_{out} refer to the number of input and output units of the layer [72]. Contrastingly, in the proposed model the *He Normal* function is used for kernel initialisation [73], as specified by the following distribution.

$$W \sim \mathcal{N}\left(0, \frac{2}{n_{in}}\right)$$

The decision to employ He Normal initialisation was motivated by the use of linear unit activation in the network. This initialisation scheme was specifically developed for ReLU and its variants, ensuring variance preservation across layers and mitigating the risk of vanishing outputs [73]. This initialisation also prevents the under-scaling of kernel weights exhibited by Glorot initialised networks, ultimately leading to improved model robustness.

L2 regularisation was applied to all convolutional kernels following an observation of extreme overfitting during the training process. The L2 regulariser serves to penalise excessively large weights that contribute little to performance improvement. Regularisation involves the addition of the L2 norm of each kernel to the loss function weighted by constant λ , following the equation

$$\text{Loss} = \text{Original Loss} + \lambda \sum_{j=1}^n w_j^2$$

where w is the array of weights in each convolutional kernel, and n the number of weights in the array. Following hyperparameter tuning, 0.1 was found to be the optimal value of λ . Although implementation of this penalty only negligibly impacted the classification accuracy the overfitting problem noticeably improved. The average training epochs until convergence was reduced by a significant margin, and the divergence of training and validation curves was reduced significantly.

Additionally, hyperparameter tuning was conducted to determine an optimal combination of values for existing hyperparameters used by EEGNet, plus the aforementioned λ . Bayesian Optimisation was employed to analyse up to 100 hyperparameter combinations and find the configuration with the lowest validation loss. Bayesian Optimisation was chosen over a grid or random search due to its efficiency in exploring high-dimensional space through selection of probable candidate solutions. The searched hyperparameters governed the rate and type of dropout layers in the model, size of convolutional kernels, number and depth of filters and degree of weight regularisation. The results of the tuning can be seen in Table 2.1.

Hyperparameter	Range of values	Step	Best Value
Dropout Rate	0.1 $\rightarrow$ 0.7	0.1	0.4
Kernel Length	8 $\rightarrow$ 64	8	32
F1 (filters)	4 $\rightarrow$ 32	4	4
D (depth of filters)	1 $\rightarrow$ 4	1	4
λ	$10^{-5} \rightarrow 0.5$	Log scale	0.1
Dropout Type	Dropout, SpatialDropout2D	Selection	Dropout

Table 2.1: Hyperparameters tuned during Bayesian optimisation

2.3.3 Training Regime

The training regime, similarly inspired by that of EEGNet, is designed to optimise classification performance while maintaining computational efficiency. The base loss function used is **categorical cross-entropy**, an appropriate and typical loss function for multi-class classifier training. The **Adam optimiser** is also used, with its adaptive learning rates facilitating stable and efficient convergence in the challenging EEG signal domain.

Although training epochs were already limited at 1000, early stopping is implemented with checkpointing to improve performance and computation time, a particularly important consideration given the complexity of the training process. Validation loss is used as a metric, with training stopped early if it does not improve for 150 epochs. A batch size of 64 is harnessed to compromise between computational efficiency and gradient stability.

A learning rate scheduler was trialled during model development but ultimately not included in the final model due to an observed detrimental effect

on classification accuracy. Regular reductions in learning rate applied to such a lightweight model were observed to induce premature convergence and ultimately hinder optimisation.

2.4 Evaluation Methods

Two methods were used to evaluate the performance of a particular model on a dataset: **Same-Subject Evaluation** and **Cross-Subject Evaluation**. These methods, the same as those used in EEGNet, were specifically chosen in order to obtain comparable results to the original paper.

Same-subject evaluation involves training the model on a single subject’s training data and testing using the subject’s evaluation set. As specified by EEGNet, stratified four-fold cross-validation is applied to the training set during this evaluation. Each subject’s training data is divided into four folds, with each fold serving as the **validation set** in one trial, with the remaining three folds forming the **training set**. In each trial, the model is trained and validated with these sets, and finally tested with the subject’s **evaluation set**, with the test accuracy recorded. This process is repeated for all 4 fold combinations for each subject. Same-subject evaluation typically yields comparatively high test accuracies, avoiding the inter-subject variability caused by head shapes, individual neurophysiology or other discrepancies [74]. Additionally, this method allows models to learn subject-specific EEG input patterns [1].

Cross-subject evaluation represents a more challenging classification task, in which the model is trained on data from all subjects except one, with the final subject reserved for testing. In each trial, one subject’s data is designated as the **test set**, while the remaining eight subjects are randomly partitioned into a **training set** of five subjects’ data and a **validation set** of the other three subjects’ data. This procedure is repeated until every subject has served as the test set once, ensuring comprehensive evaluation across participants. Similarly to same-subject evaluation, the test accuracies of all subjects are recorded. Cross-subject classification accuracies are generally lower, in part due to the variability in production of MI signals between individuals, leading to domain shifts that hinder generalisation [74].

All evaluations are conducted in a Python 3.8 environment with CUDA support, running on an Ubuntu workstation equipped with dual NVIDIA K40 GPUs.

CHAPTER 3

Results & Discussion

In this section, the performance of the outlined model is examined in detail. All reported results correspond to the best performance obtained over 10 independent runs for each model and evaluation setting. For comparison, a selection of relevant peer models have also been implemented and evaluated, with the performance of the following four models considered:

- Shallow ConvNet
- Deep ConvNet
- EEGNet 4,2
- EEGNet 8,2

These specific models were selected as they not only inspired the design of the proposed model, but also represent some of the most widely used CNN-based baselines for EEG classification, with both light and heavy architectures covered. Their inclusion enables a comprehensive and meaningful comparison against the proposed model. Each of these existing models were reproduced according to their original publications (aside from some minor modifications to ConvNet models¹), including the associated pre-processing pipelines, network architectures, hyperparameter values, and optimisation strategies. Detailed descriptions of each of these baselines are provided in Section 1.2. The models were all implemented using Tensorflow’s Keras API in Python.

¹Max normalisation was applied to convolutional blocks, and the default parameters of batch normalisation layers were adjusted. The number of filters, convolutional stride and pooling size in Shallow ConvNet were also modified. These changes were informed by direct advice from the original model authors as a strategy to improve classification accuracy [75].

3.1 Same-Subject Evaluation

The first evaluation method involved assessing the proposed model’s classification accuracy on same-subject data from Datasets 2a and 2b.

3.1.1 Dataset 2a

Classification accuracy over the nine subjects of Dataset 2a was assessed first. The subject-wise and average accuracies for each model is seen in Table 3.1. The proposed model demonstrated consistently strong performance relative to the baseline models, achieving the highest classification accuracy for eight out of nine subjects. Subject A09 was the exception, where EEGNet 8,2 outperformed the proposed model by 2.43%. The proposed model featured the highest average classification accuracy across all subjects, with an average of **65.70%**. This showed an improvement of 10.10% and 6.30% from EEGNet 4,2 and 8,2 models respectively. This is notable outcome, as the proposed pipeline builds directly upon these two models, demonstrating the impact of introduced architectural and training enhancements. The performance gain is observed across subjects, showing the robustness of these changes to inter-subject variability and validating them beyond chance-based fluctuation.

Model	Subject									Avg
	A01	A02	A03	A04	A05	A06	A07	A08	A09	
Shallow ConvNet	62.76	40.10	77.26	57.03	50.26	41.06	81.51	70.23	73.96	61.57
Deep ConvNet	54.95	26.56	65.54	27.60	25.95	26.22	26.13	52.43	66.32	41.30
EEGNet 4,2	72.48	38.19	74.13	42.62	30.03	33.16	58.59	72.66	78.47	55.60
EEGNet 8,2	71.70	29.08	77.34	41.41	42.45	39.84	79.25	74.13	79.43	59.40
Proposed Model	79.86	42.36	83.68	57.03	58.16	44.53	81.51	74.13	77.00	65.70

Table 3.1: Same-subject results on Dataset 2a (Accuracy %).

Shallow ConvNet achieved a competitive average accuracy of 61.57%, outperforming both EEGNet models, and even achieving the same accuracy as the proposed model for subjects A04 and A07. Meanwhile, Deep ConvNet performed remarkably poorly with an average accuracy of just 41.30%. This significant performance gap indicates that greater architectural complexity does not necessarily lead to improved performance.

It is evident that some subjects in this dataset present a more intrinsically challenging classification task than others, with subjects A02, A04, A05 and A06 exhibiting consistently low accuracies across all evaluated models. Several factors

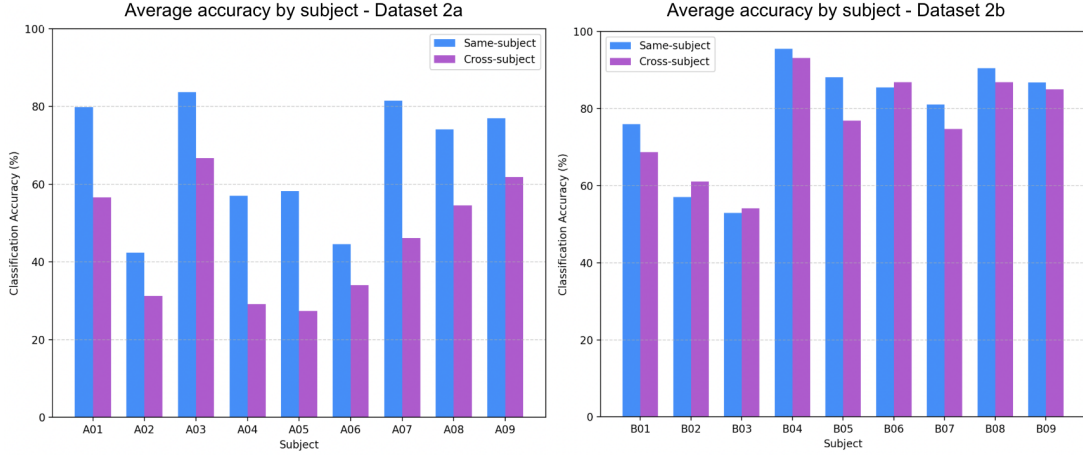


Figure 3.1: Average classification accuracies of the proposed model by subject. Results are shown from Dataset 2a (left) and Dataset 2b (right).

likely account for this, including low SNRs in recordings, atypical neurophysiological patterns, or simply poor ability to perform clear motor imagery. Notably, these were also the subjects in which the proposed model recorded the largest relative and absolute performance gains, achieving average accuracy improvements of **14.52%** and **12.33%** from the EEGNet 4,2 and 8,2 models, respectively. This included a remarkable **93.67%** relative increase in accuracy of subject A05, from 30.03% in EEGNet 4,2 to nearly 60% accuracy in the proposed model. The subject-wise accuracies of the proposed model’s classification are visualised in Figure 3.1, displaying the contrast in average accuracies of each subject.

3.1.2 Dataset 2b

Next, same-subject classification was undertaken using the subjects from Dataset 2b. As previously noted, less emphasis was placed on optimising performance of this dataset compared to Dataset 2a. Hyperparameter tuning was carried out using Dataset 2a, and several design choices were specifically tailored to accommodate a larger number of output classes. Consequently, the comparative results on Dataset 2b are unsurprisingly less favourable.

As displayed in Table 3.2, the classification accuracies of the proposed model over this dataset are competitive, but not the comparative best. The proposed model achieves the highest accuracy in the classification of just two subjects: B01 and B02. For the remaining seven subjects, the proposed model achieves accuracies slightly below those of the EEGNet models, trailing the top-performing

Model	Subject									Avg
	B01	B02	B03	B04	B05	B06	B07	B08	B09	
Shallow ConvNet	68.52	51.79	55.16	95.23	82.89	72.34	75.00	90.94	81.88	74.86
Deep ConvNet	58.59	50.00	57.42	91.95	70.47	59.38	51.17	79.45	76.09	66.06
EEGNet 4,2	70.00	53.66	61.64	97.73	88.98	84.53	81.33	92.34	88.13	79.80
EEGNet 8,2	69.37	54.82	62.42	97.58	91.95	85.62	82.19	92.42	86.80	80.40
Proposed Model	75.94	57.05	52.97	95.47	88.20	85.55	81.09	90.55	86.80	79.03

Table 3.2: Same-subject results on Dataset 2b (Accuracy %).

model by less than 4% in six of the seven participants. As a result, the two highest average accuracies for this evaluation are EEGNet 8,2 and 4,2, with scores of 80.40% and 79.80% respectively. The proposed model follows just behind with **79.03%** average accuracy. Shallow ConvNet does not perform as well on this dataset, achieving 74.86% average accuracy, while Deep ConvNet once again performs poorly with just 66.06%.

In Dataset 2b, subjects B01, B02 and B03 can be considered difficult to classify, as supported by subjectwise results and Figure 3.1. The proposed model’s performance on these challenging subjects is somewhat strong, achieving its two best relative performances on B01 and B02. However, this pattern does not hold for subject B03, where the proposed model achieves its lowest accuracy, scoring nearly 10% below the top-performing EEGNet 8,2.

3.2 Cross-Subject Evaluation

Cross-subject evaluation represents a substantially different classification domain, with the model trained and evaluated on separate subjects.

3.2.1 Dataset 2a

The proposed model again performs strongly on classification of Dataset 2a, achieving the highest average classification accuracy of **45.29%**. EEGNet 4,2 and 8,2 are not far behind, recording averages of 44.14% and 42.63% respectively. The performance gap between these models and the proposed model is noticeably smaller, but still relevant. The Shallow and Deep ConvNet models again lag behind, recording the two worst average accuracies, further demonstrating their ineffectivity for this task. The performances of all models on Dataset 2a take a substantial dip from same-subject performances. This was an expected result, as cross-subject classification is an inherently more challenging task.

The results of the proposed model are less consistent across subjects in this evaluation. The model attains the highest accuracy for four of the nine subjects, outperforming the next best model by an average margin of 6.2%, a notable gap given the generally low accuracy levels. However, its performance varies considerably across subjects, with the three weakest results falling over 11% below the leading models. The model even records a worse accuracy than all four baseline models on subject A04. These inconsistencies can be partly attributed to the difficulty of cross-subject classification in a four-class MI dataset. Additionally, the cross-subject evaluation paradigm itself also contributes to this inconsistency, as it relies on a subject-wise random split to form training and validation sets. As previously discussed and illustrated in Figure 3.1, there is substantial variability in the classification difficulty between subjects. As a result, this variability is carried into model performance, with different subject combinations in each split leading to substantial differences in test set accuracy.

Model	Subject									Avg
	A01	A02	A03	A04	A05	A06	A07	A08	A09	
Shallow ConvNet	54.51	28.12	57.64	42.36	26.74	32.29	34.72	50.00	43.75	41.33
Deep ConvNet	29.17	23.61	53.12	38.19	27.78	29.51	32.64	51.74	48.26	37.11
EEGNet 4,2	65.62	26.74	56.94	39.58	23.96	32.99	38.19	65.62	47.57	44.14
EEGNet 8,2	63.19	26.39	68.75	30.21	25.69	24.65	37.15	58.33	49.31	42.63
Proposed Model	56.60	31.25	66.67	29.17	27.43	34.03	46.18	54.51	61.81	45.29

Table 3.3: Cross-subject results on Dataset 2a (Accuracy %).

3.2.2 Dataset 2b

The final evaluation involved cross-subject analysis of Dataset 2b. The comparative results of each model were observed to be very similar to the same-subject evaluation of this dataset: with EEGNet 8,2 again achieving the highest average accuracy of 78.49%. Similarly, it was closely followed by EEGNet 4,2 (77.79%) and the proposed model (**76.37%**), with all three models separated by just over 2% accuracy. Shallow and Deep ConvNets again perform poorly, with 72.54% and 74.00% averages respectively. The proposed model does not record the best accuracy in any of the nine subjects, a surprising result given its competitive average accuracy. This suggests that while the model exhibits solid consistency, it lacks peaks in this classification which favour specific subjects. This is hypothesised to indicate that the proposed model’s feature extraction over this dataset is more globally balanced, but less able to exploit subject-specific neural characteristics.

Another surprising aspect of these results is observed when comparing the accuracies to the same-subject results of the same dataset, as only a very marginal performance dip is observed overall. The proposed model, both EEGNet variants and Shallow ConvNet all experience a mere $\sim 2\%$ reduction in average accuracy. Contrastingly, the reduction in average accuracy from same-subject to cross-subject analysis on Dataset 2a is 15-20% for each model. As shown by Figure 3.1, in subject B02 and B03 and B06, the proposed model’s cross-subject accuracy remarkably even **outperforms** the same-subject accuracy. The unexpected stability in cross-subject performance on Dataset 2b is due to the set’s comparatively low inter-subject variability. This can be attributed to several factors, discussed in more detail in Section 3.3.

Model	Subject									Avg
	B01	B02	B03	B04	B05	B06	B07	B08	B09	
Shallow ConvNet	61.56	50.71	56.25	93.44	68.13	75.31	71.25	91.25	85.00	72.54
Deep ConvNet	61.56	62.86	59.06	90.00	71.88	74.06	71.25	93.44	81.87	74.00
EEGNet 4,2	67.81	61.43	52.50	96.56	84.69	84.69	77.50	91.25	84.69	77.79
EEGNet 8,2	69.69	58.93	60.62	96.25	79.37	88.12	79.06	87.50	86.88	78.49
Proposed Model	68.75	61.07	54.06	93.13	76.88	86.88	74.69	86.88	85.00	76.37

Table 3.4: Cross-subject results on Dataset 2b (Accuracy %).

3.3 Comparison of Datasets

Although average accuracies are substantially higher for every model on Dataset 2b than Dataset 2a, this should not alone be interpreted as evidence of superior performance. The classification task of Dataset 2b is inherently less complex, with only two output classes rather than the four in Dataset 2a, naturally leading to inflated accuracy. A better basis for comparison is the *Cohen’s Kappa* (κ) value of each evaluation, defined as

$$\kappa = \frac{p_o - p_e}{1 - p_e},$$

where p_o refers to the observed accuracy, and p_e the expected accuracy by chance (i.e. $1/N$). The proposed model achieves $\kappa = 0.543$ for 2a and $\kappa = 0.581$ for 2b in same-subject classification, and $\kappa = 0.271$ for 2a and $\kappa = 0.527$ for 2b in cross-subject classification. By this metric, the model’s classification on Dataset 2b is more discriminative across the board, despite its poorer performance on this dataset relative to peer models. This discrepancy reflects significant differences in the quality of MI signals between the two datasets. This is likely due to the online trials of Dataset 2b, encouraging subjects to produce more stereotyped MI

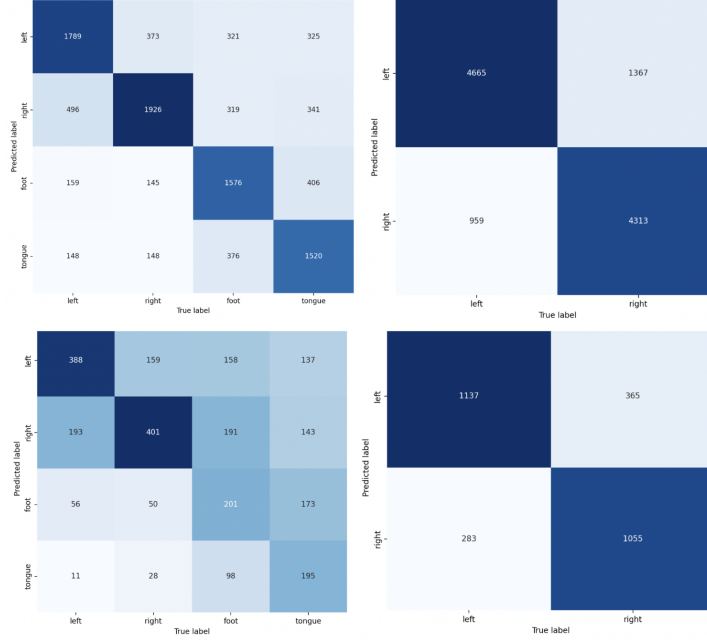


Figure 3.2: Confusion matrices for same-subject evaluation of Datasets 2a (top left) and 2b (top right), and cross-subject evaluation of Datasets 2a (bottom left) and 2b (bottom right).

as they are driven to replicate the imagery which produces positive feedback. The resulting MI has less variability between subjects, allowing for more consistent classification, particularly in the cross-subject domain.

The confusion matrices in Figures 3.2 are another means of direct comparison of both datasets' results. A clear difference in recall values between MI classes is seen in Dataset 2a, with this contrast particularly evident when comparing left/right hand MI to tongue/foot MI in both same-subject and cross-subject evaluation. Both hand classes exhibit comparatively higher recalls, suggesting that hand MI may be more reliably and easily classified. This observation aligns with findings in literature, with Pfurtscheller et al. reporting that hand MI consistently induces significant μ rhythm ERD across all subjects, whereas foot and tongue MI typically elicit weaker μ rhythm ERS in only a subset of subjects, thereby exhibiting weaker discriminability [76]. This helps to explain the higher κ scores of Dataset 2b, as this dataset only features the comparatively discriminable left and right hand MI classes, and not the trickier tongue and foot classes.

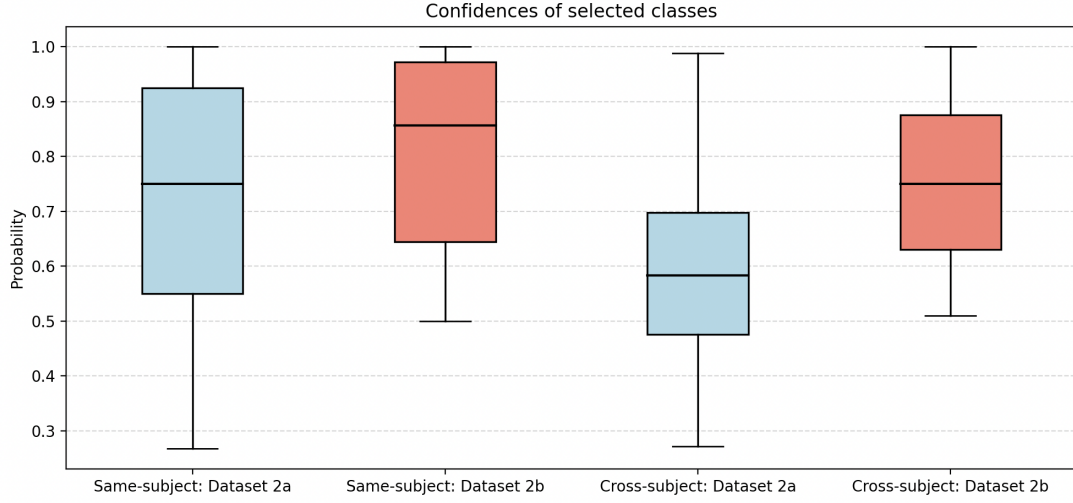


Figure 3.3: The distribution of top-1 class confidences for evaluations of Datasets 2a and 2b. The confidence value represents the expected likelihood of a trial belonging to a specific class.

Figure 3.3 displays the distribution of confidence values of selected classes, a chart further supporting this sentiment. Dataset 2a exhibits a median top-1 confidence of 0.750 in same-subject and 0.583 in cross-subject classification. Dataset 2b records higher medians across each evaluation with values of 0.857 and 0.751 respectively. This indicates that model predictions on Dataset 2b are significantly more confident on average, reflecting a clearer class separation and supporting the idea of this set having more discriminable MI classes. However, a significant portion of this disparity can also be attributed to the set having just two target classes rather than four, and subsequently having top-1 prediction confidences begin at 50% rather than 25%.

3.4 Ablation Study

Two additional ablation studies were conducted on the proposed model. These involved selectively removing components of the pipeline to assess their individual impact on performance, with one trial assessing the novel pre-processing steps, and the other the new network design. Same-subject evaluation was chosen for this study due to its stability, as the subject-wise random training/validation splits used in cross-subject evaluation introduce variability that can mask the effects of model or pre-processing changes. This controlled setting makes isolated effects clearer.

The first two additional models assessed are EEGNet 4,2 and 8,2 with the inclusion of the newly introduced ICA and data augmentation pre-processing steps. The performance of these modified models was directly compared to their original counterparts, with the inclusion of ICA and augmentation intentionally isolated to assess the individual contribution of these techniques to overall performance. Figure 3.4 depicts this comparison, revealing clear accuracy improvements in both EEGNet models following the addition of these steps to the pipeline. EEGNet 4,2 has its accuracy slightly improved by 1.62%, while 8,2 experiences a larger 5.22% jump. Additionally, the epoch corresponding to the model’s best performance is reported as a metric, providing insight into the speed at which training converges. The figure shows a significant reduction in the number of training epochs required for both EEGNet models, a desirable result which further highlights the benefits of incorporating ICA and data augmentation. This result not only reflects improved training efficiency, but also suggests a more generalisable and less overfit model is produced, as the model is able to learn more discriminative representations earlier. However, it is important to note that the reduction in epochs is also influenced by the increased training set size resulting from data augmentation. Consequently, while the number of epochs required for convergence decreases, the total computational workload may not be reduced proportionally, due to the greater number of training samples in each forward pass.

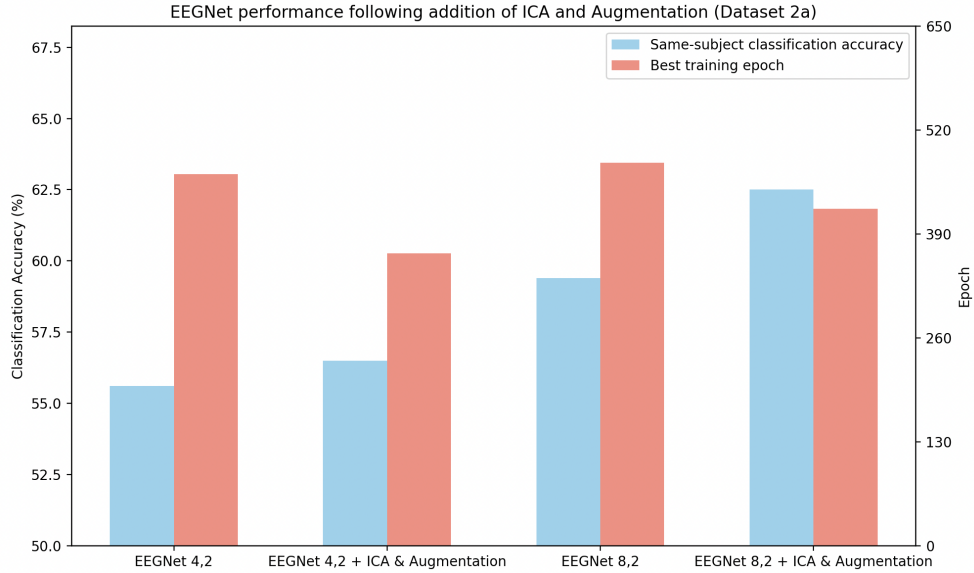


Figure 3.4: Average same-subject classification accuracy on Dataset 2a of EEG-Net models with and without novel pre-processing techniques.

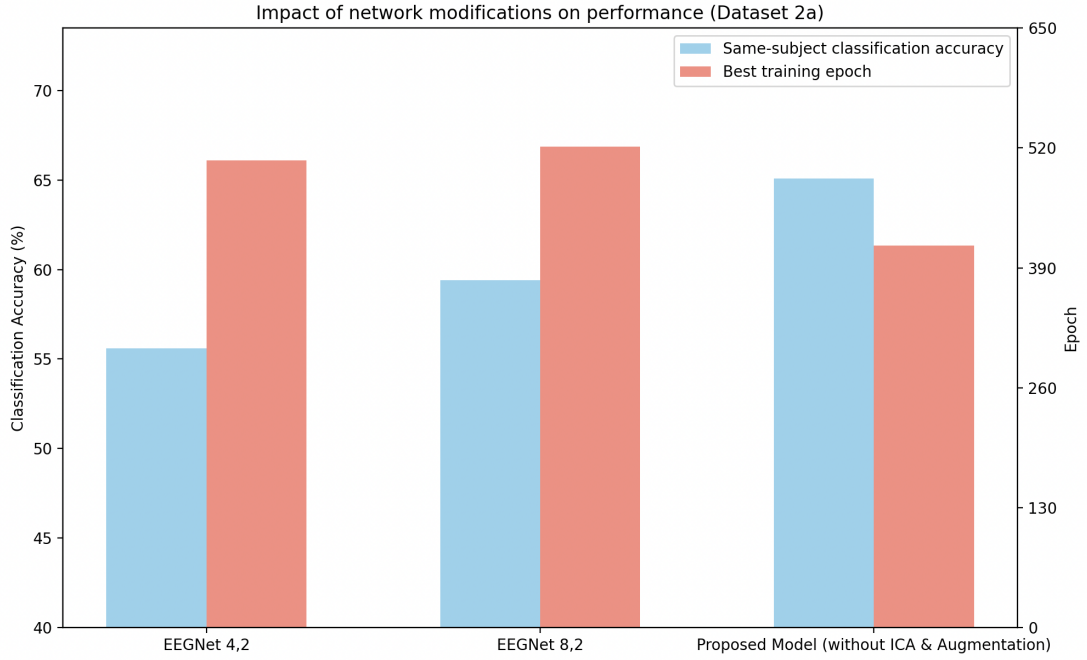


Figure 3.5: The impact of differing model architectures on performance, with frozen pre-processing pipelines.

One final modified model was evaluated on this dataset: the proposed model’s CNN with EEGNet’s pre-processing pipeline, (i.e. not featuring any ICA or augmentation). In contrast to the previous comparison, which examined the effect of adding pre-processing steps to the EEGNet models, this experiment isolates the contribution of the network architecture itself by comparing EEGNet directly against the proposed model under identical pre-processing conditions. Visualised in Figure 3.5, the modified model significantly outperformed both EEGNet variants in average accuracy, demonstrating the positive impact of network modifications. Some of the major contributing changes are the additional dense network layer, the addition of weight regularisation, modified kernel initialisation and use of tuned hyperparameter values. Similarly to the previous experiment, the epoch number also dramatically reduces from that of both EEGNet models. This is an encouraging result, suggesting that the proposed network promotes significantly more generalisability and more efficient learning of representations. Importantly, this reduction in epoch number cannot be attributed to an increased training set size, as this comparison features the same pre-processing pipeline and inputted data.

3.5 Limitations & Future Work

While the project produced valuable and insightful results, it was subject to several notable limitations. These arose from time and resource constraints, as well as experimental choices, which may have impacted the overall quality of the results.

3.5.1 Replication of Results

Some of the originally reported results from baseline models could not be fully replicated in this research. Same-subject accuracies of both EEGNet and ConvNet models on Dataset 2a were reported in the original EEGNet paper [1], which found Shallow ConvNet achieving the highest average accuracy, followed by EEGNet 8,2 and EEGNet 4,2, with Deep ConvNet performing worst by a large margin. This comparative performance was mirrored in our study, as seen in Table 3.1. However, despite faithfully replicating all original pre-processing pipelines, model architectures and evaluation methodology as reported, the reported average accuracies were up to 10% higher than those achieved in our research. This discrepancy likely reflects differences in implementation environments or even subtle variations in pre-processing pipelines, optimisations and model design that were not fully documented. It may also represent the inherent variability in deep learning model performance on EEG data, where small changes can lead to disproportionately large shifts in accuracy. The paper also reports cross-subject accuracies for the same four models on this dataset, results which **were** replicated by our research. All collected accuracies fell within the reported *2 standard error* range of the scores in this paper, differences explained by normal sampling variability.

3.5.2 Resource Constraints

If time and resources had permitted, a broader range of both peer and specialised state of the art models could have been replicated and evaluated, creating a more comprehensive comparison of architectures and pipelines. The benchmarking could have been extended to include transformer-based architectures like CTNet [50], Riemannian geometry based methods such as XDAWN + RG [1], or hybrid deep learning models, in order to provide deeper insights into relative strengths and weaknesses. Furthermore, the model’s performance, particularly with regards to its generalisability, could be more rigorously assessed with the evaluation against additional independent datasets. In future work, more

datasets, preferably those with differing numbers of MI classes, recording conditions and subject attributes, should be included for a more robust evaluation, ultimately strengthening the conclusions of the study. Finally, access to greater computational resources could have permitted a more comprehensive and precise hyperparameter tuning as well as longer training processes.

3.5.3 Online Implementation

For future deployment in an online BCI setting where movement commands must be processed in real time, further development and adaptation of this model is required. First, the pre-processing pipeline would have to process inputted signals with extremely low latency to facilitate responsive device control. Current implementations of some steps, particularly artefact removal, would need reworking to meet these latency requirements. Input handling modifications would also be required, with a means of converting continuous inputs into fixed-length sliding windows integrated into the pipeline. In addition to these model changes, the entire system would have to be integrated into a closed-loop BCI framework with real-time communication with both signal acquisition hardware and the external movement device. For successful integration into a BCI system, a key design decision would involve choosing between a per-subject calibration approach, analogous to same-subject evaluation, or pre-training models for immediate use, following a cross-subject evaluation paradigm.

Although strong compared to peer models, the average accuracies achieved in both same-subject and cross-subject evaluation are still below the thresholds required for any realistic implementation in a medical context. Future work should therefore focus not only on improving the model’s processing capabilities and implementation, but also on substantially increasing its classification accuracy to meet the stringent standards of clinical BCI applications.

CHAPTER 4

Conclusion

In this project, several valuable contributions to the fields of EEG classification and BCI research are made. A robust pre-processing pipeline for signal preparation is developed, incorporating novel implementations of ICA and data augmentation techniques, leading to improved classification and training efficiency. An EEGNet-based convolutional neural network architecture for end-to-end EEG-MI classification is introduced. This architecture integrates targeted enhancements such as improved weight initialisation, regularisation, tuned hyperparameters, and an additional network block to boost feature extraction and generalisability. Comprehensive evaluations are conducted using two benchmark datasets, enabling fair comparisons with four established baseline models. Finally, ablation studies are performed to isolate and quantify the contributions of individual modifications, demonstrating the positive impact of both pre-processing and architectural changes.

The proposed model achieves strong classification performance when compared to baseline architectures, outperforming all assessed baselines on Dataset 2a, and performing competitively on Dataset 2b. These improvements result from targeted enhancements to EEGNet, which deliver measurable accuracy gains without substantially increasing model complexity. **The findings therefore validate the research hypothesis in Section 1.4.1**, showing that focused refinements to baseline models can meaningfully improve both same-subject and cross-subject accuracy in EEG classification systems. Overall, this research presents meaningful advancements to EEG-based classification, while highlighting the remaining challenges which must be addressed to enable deployment in real-world BCI applications.

Appendix 1: Comparison of Networks

	Proposed Model		EEGNet (4,2)	
Layer	Parameter	Size	Parameter	Size
Input	–	(C, T)	–	(C, T)
Block 1				
Temporal Convolution	Filters: 4	$(4, C, T)$	Filters: 4	$(4, C, T)$
Batch Norm	Size: (1,32)		Size: (1,32)	
Depthwise Convolution	–	$(4, C, T)$	–	$(4, C, T)$
Batch Norm	Filters: 16	$(16, 1, T)$	Filters: 16	$(8, 1, T)$
ELU Activation	Size: (C,1)		Size: (C,1)	
Avg Pooling	–	$(16, 1, T)$	–	$(8, 1, T)$
Dropout	–	$(16, 1, T)$	–	$(8, 1, T)$
	Size: (1,4)	$(16, 1, T//4)$	Size: (1,4)	$(8, 1, T//4)$
	Rate: 0.4	$(16, 1, T//4)$	Rate: 0.25/0.5 ¹	$(8, 1, T//4)$
Block 2				
Separable Convolution	Filters: 16	$(16, 1, T//4)$	Filters: 8	$(8, 1, T//4)$
Batch Norm	Size: (1,16)		Size: (1,16)	
ELU Activation	–	$(16, 1, T//4)$	–	$(8, 1, T//4)$
Avg Pooling	–	$(16, 1, T//4)$	–	$(8, 1, T//4)$
Dropout	Size: (1,8)	$(16, 1, T//32)$	Size: (1,8)	$(8, 1, T//32)$
	Rate: 0.4	$(16, 1, T//32)$	Rate: 0.25/0.5 ¹	$(8, 1, T//32)$
Block 3				
Global Avg Pooling	–	(16)		
Dense	Neurons: 64	(64)		
Leaky ReLU Activation	–	(64)		
Batch Norm	–	(64)		
Dropout	Rate: 0.4	(64)		
Classification Head				
Flatten	–	(64)	–	(64)
Dense	Neurons: N	(N)	Neurons: N	(N)
Softmax	–	(N)	–	(N)
Activation				

Table 1: Side-by-side comparison of the proposed model architecture and the EEGNet (4,2) architecture. C refers to the number of EEG channels, T refers to the number of samples per trial and N refers to the number of output classes.

¹0.5 is used for same-subject evaluation, while 0.25 is used for cross-subject evaluation

Appendix 2: AI Declaration

During the development of this dissertation, I used AI (in particular ChatGPT) to assist in the generation of some visuals, including the code for Python-based charts and graphs, and the generation and tweaking of some LaTeX tables and equations. For all of these, I obtained the data and values from my own research, simply using AI as a visualisation tool. I also used AI to assist with some LaTeX compilation errors and document debugging. The purpose of this was to reduce my workload, in particular by reducing the time spent on tedious tasks such as graph formatting.

Appendix 3: Code Availability

The full repository containing the proposed model implementation and evaluation scripts can be found at: www.github.com/b0b3000/EEG_MI_Classification

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